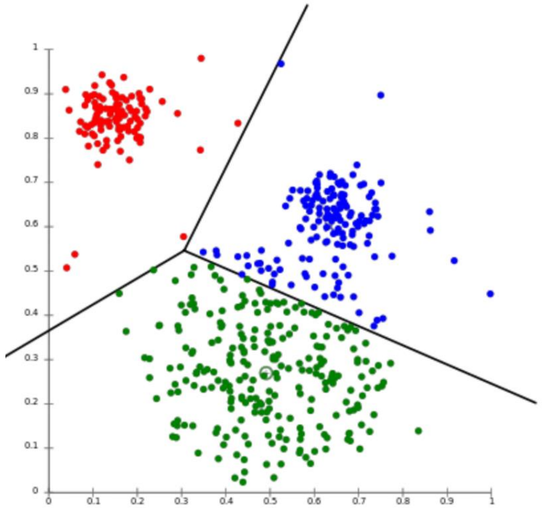
DATA

k-means

D data & k sets $\mathbb{S} = \{\mathcal{S}_1, \cdots, \mathcal{S}_k\}$

$$\arg\min_{\mathbb{S}} \sum_{i=1}^{k} \sum_{j \in \mathcal{S}_i} \|\mathbf{x}^j - \mu^i\|^2 \quad \text{with} \quad \mu^i = \frac{1}{\mathrm{S}_i} \sum_{j \in \mathcal{S}_i} \mathbf{x}^j$$

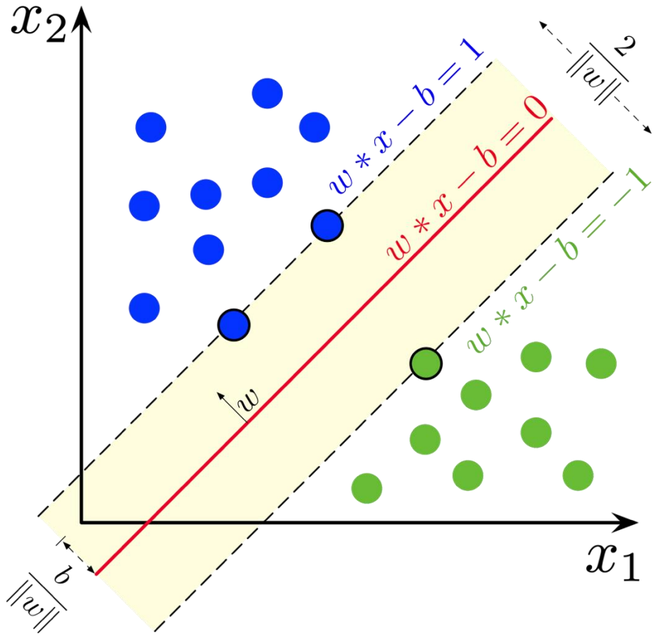
being S_i the number of data in the set $\mathcal{S}_i$



Classification: SVM

Support Vector Machine

Data $\left\{ \begin{array}{l} \mathbf{x}^i \\ \mathcal{O}^i \end{array} \right.$ with $\mathcal{O}^i = \pm 1$



Linear SVM

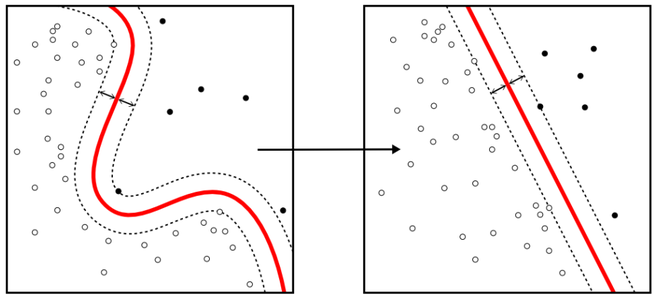
Minimize $\|\mathbf{w}\|$

with $\mathcal{O}^i(\mathbf{w} \cdot \mathbf{x}^i - b) \geq 1$

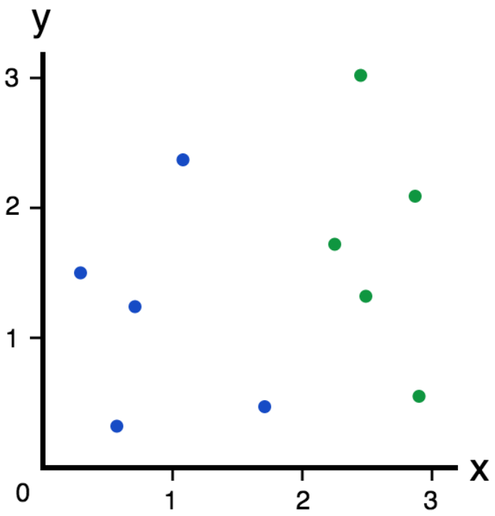
Classifier: $\mathtt{sgn}(\mathbf{w} \cdot \mathbf{x} - b)$

Nonlinear SVM

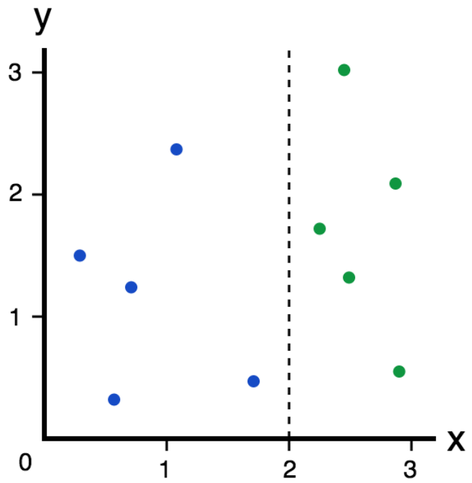
The scalar product is replace by the so-called kernel trick: $\kappa(\mathbf{x}^i, \mathbf{x}^j)$



Classification: Decision Trees

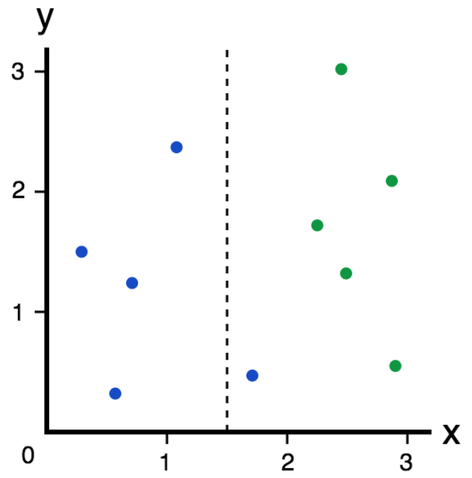


The Dataset

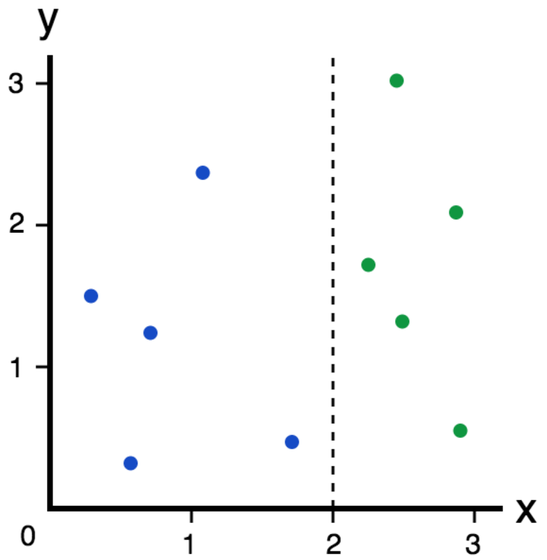


A Perfect Split

$$G = \sum_{i=1}^{C} p(i) * (1 - p(i))$$



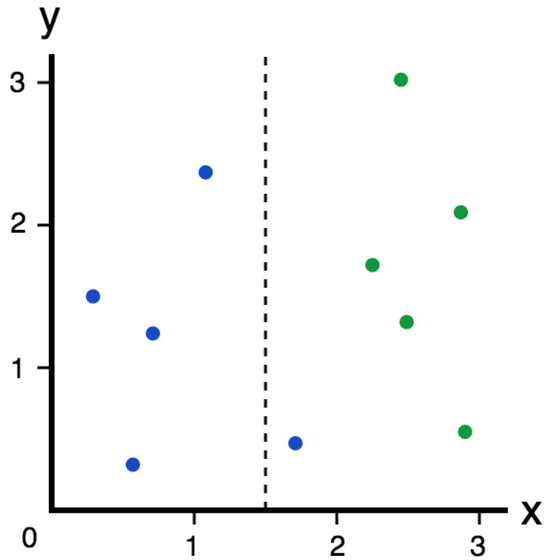
An Imperfect Split



A Perfect Split

$$G_{left} = 1 * (1 - 1) + 0 * (1 - 0) = \boxed{0}$$

$$G_{right} = 0 * (1 - 0) + 1 * (1 - 1) = \boxed{0}$$

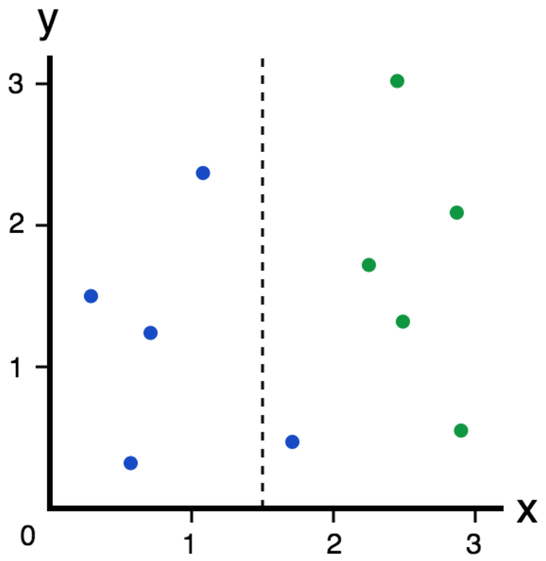


An Imperfect Split

$$G_{left} = \boxed{0}.$$

$$\begin{aligned} G_{right} &= \frac{1}{6} * (1 - \frac{1}{6}) + \frac{5}{6} * (1 - \frac{5}{6}) \\ &= \frac{5}{18} \\ &= \boxed{0.278} \end{aligned}$$

Picking The Best Split



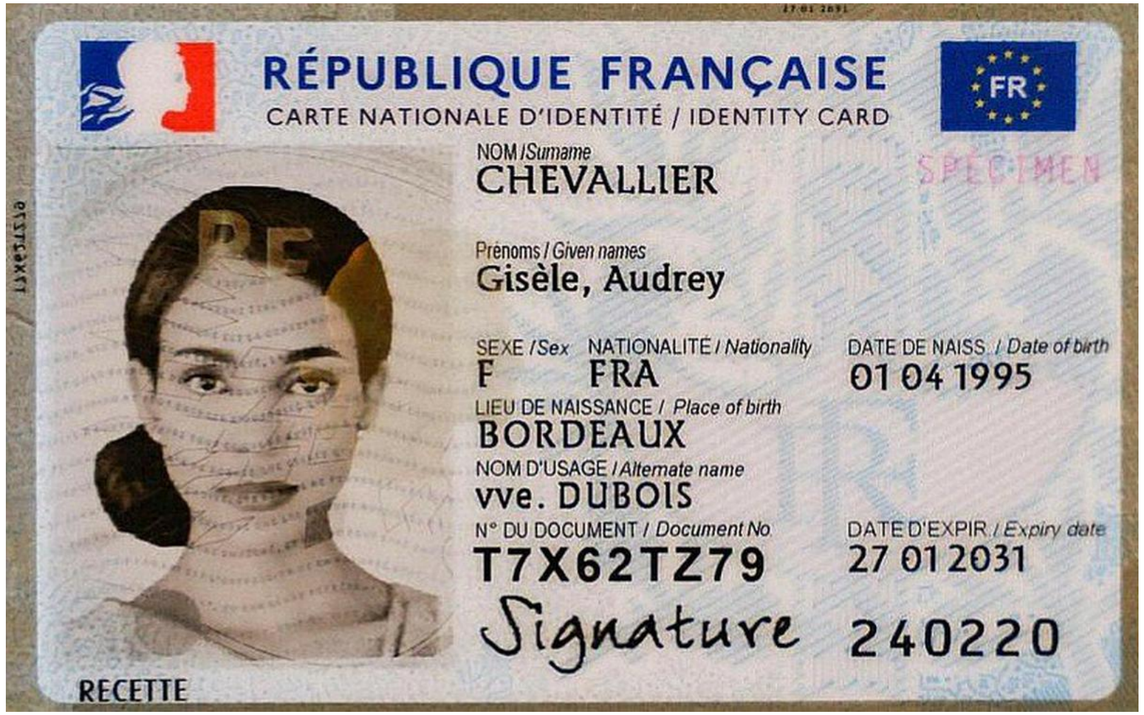
An Imperfect Split

- Before the split (the entire dataset): 0.5
- Left Branch: 0
- Right Branch: 0.278

$$(0.4 * 0) + (0.6 * 0.278) = 0.167$$

$$0.5 - 0.167 = \boxed{0.333}$$

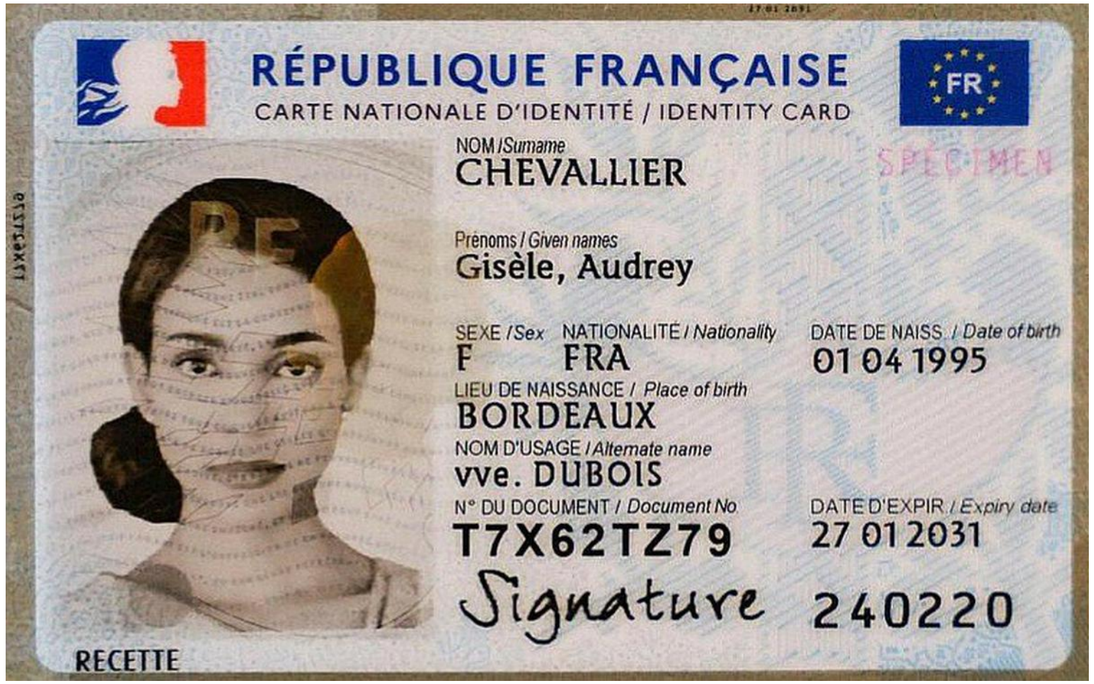
Gini Gain of the perfect split on our dataset is $0.5 > 0.333$.



A data contains multiple heterogeneous information

Data


RÉPUBLIQUE FRANÇAISE
CARTE NATIONALE D'IDENTITÉ / IDENTITY CARD
FR
NOM / Surname
MARTIN
SPÉCIMEN
Prénoms / Given names
Maëlis-Gaëlle, Marie
SEXE / Sex
F
NATIONALITÉ / Nationality
FRA
DATE DE NAISS. / Date of birth
13 07 1990
LIEU DE NAISSANCE / Place of birth
PARIS
NOM D'USAGE / Alternate name
usage NOM D'USAGE
N° DU DOCUMENT / Document No.
D2H6862M2
DATE D'EXPIR. / Expiry date
11 02 2030
Signature
546497


RÉPUBLIQUE FRANÇAISE
CARTE NATIONALE D'IDENTITÉ / IDENTITY CARD
FR
NOM / Surname
CHEVALLIER
SPÉCIMEN
Prénoms / Given names
Gisèle, Audrey
SEXE / Sex
F
NATIONALITÉ / Nationality
FRA
DATE DE NAISS. / Date of birth
01 04 1995
LIEU DE NAISSANCE / Place of birth
BORDEAUX
NOM D'USAGE / Alternate name
vve. DUBOIS
N° DU DOCUMENT / Document No.
T7X62TZ79
DATE D'EXPIR. / Expiry date
27 01 2031
Signature
240220
RECETTE

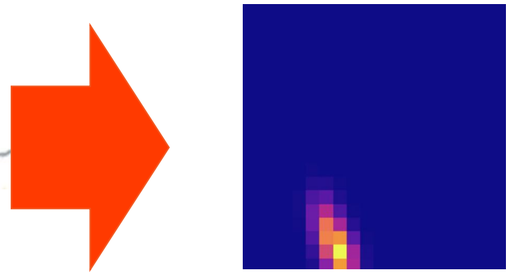

0 1 2 3 4 5 6 7 8 9 10

TDA

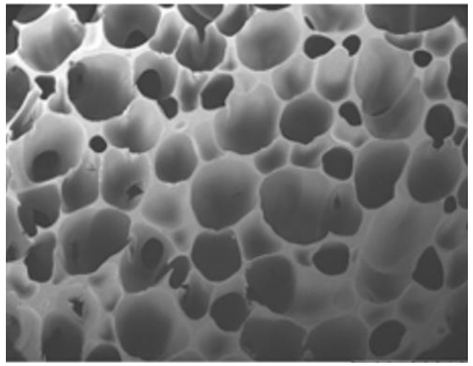


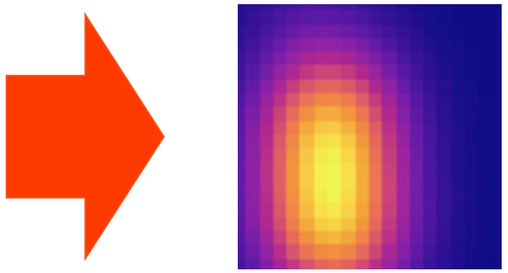
Time series





Complex images





A sort of goal oriented

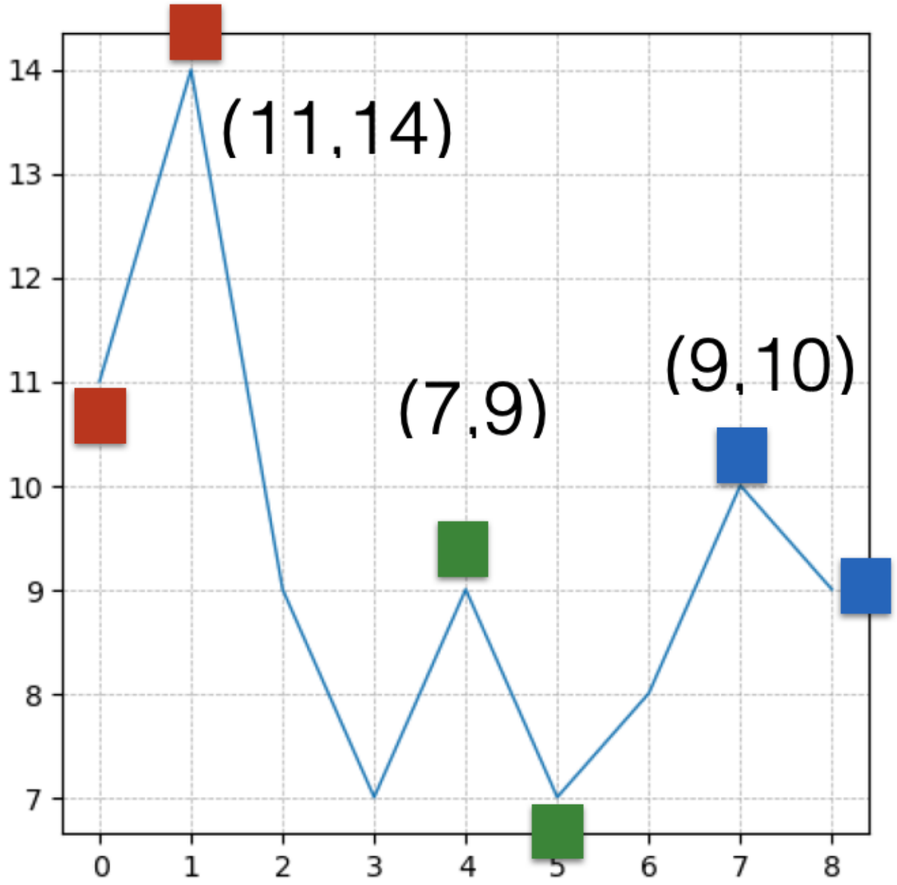
QR code / Passeport



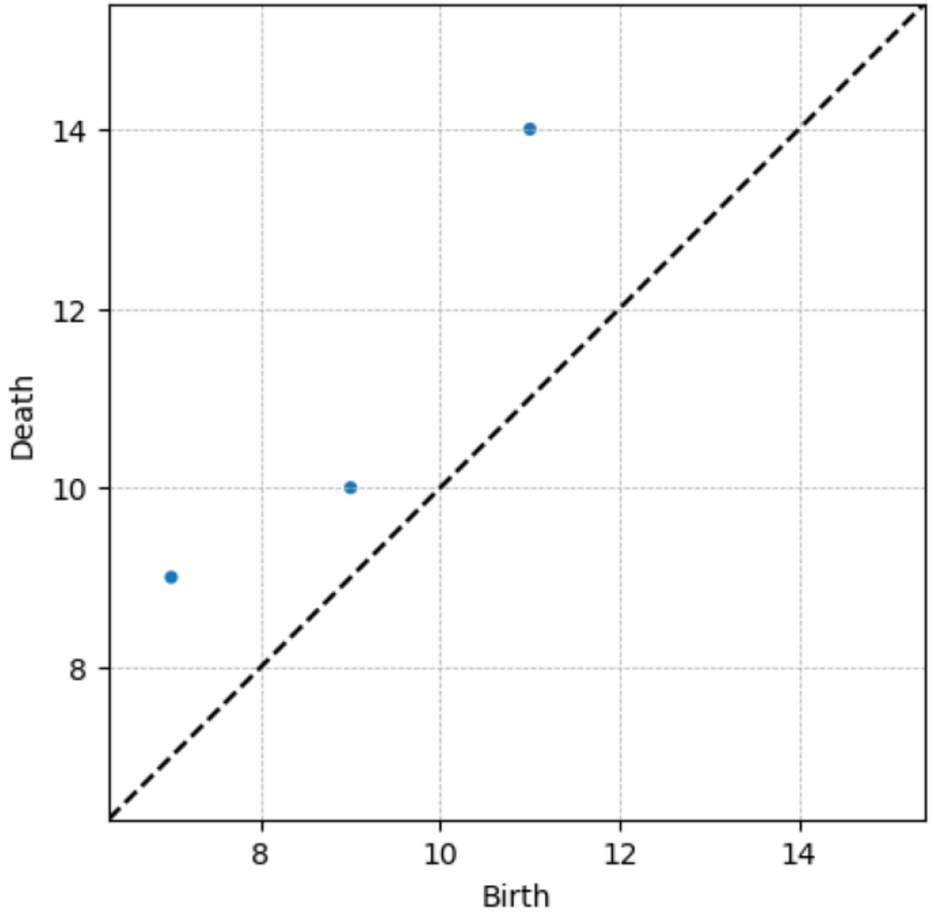
BUT in a vector space

TDA – Topological Data Analysis - applied to Time series

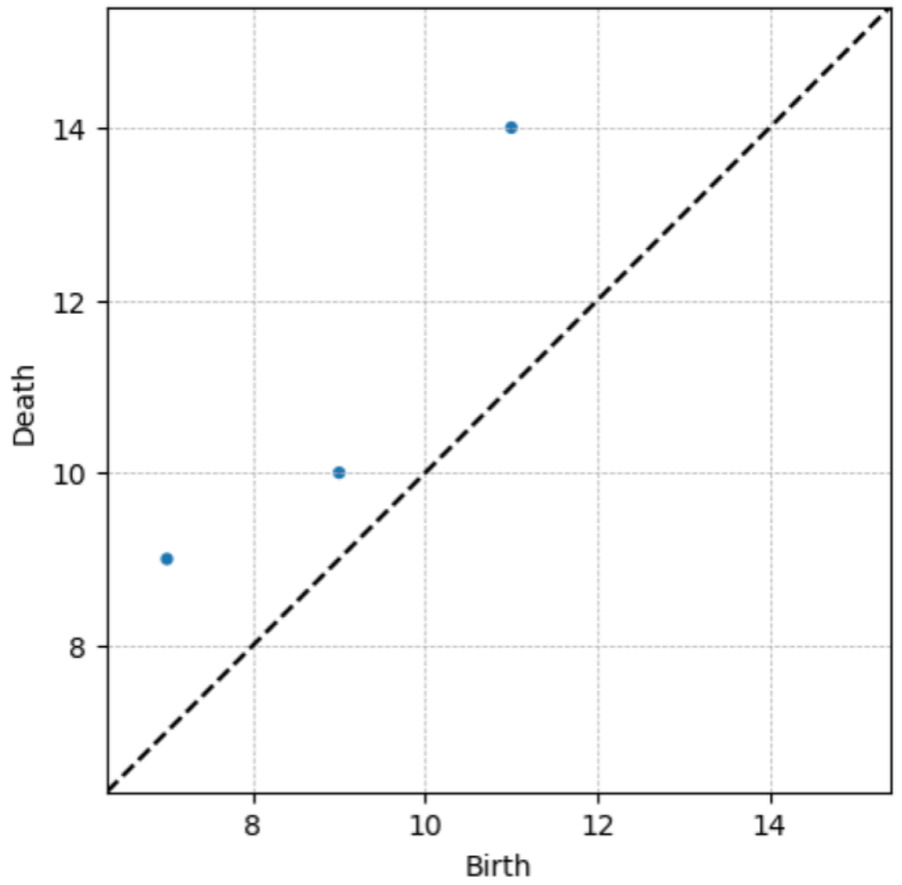
Pairing min-max



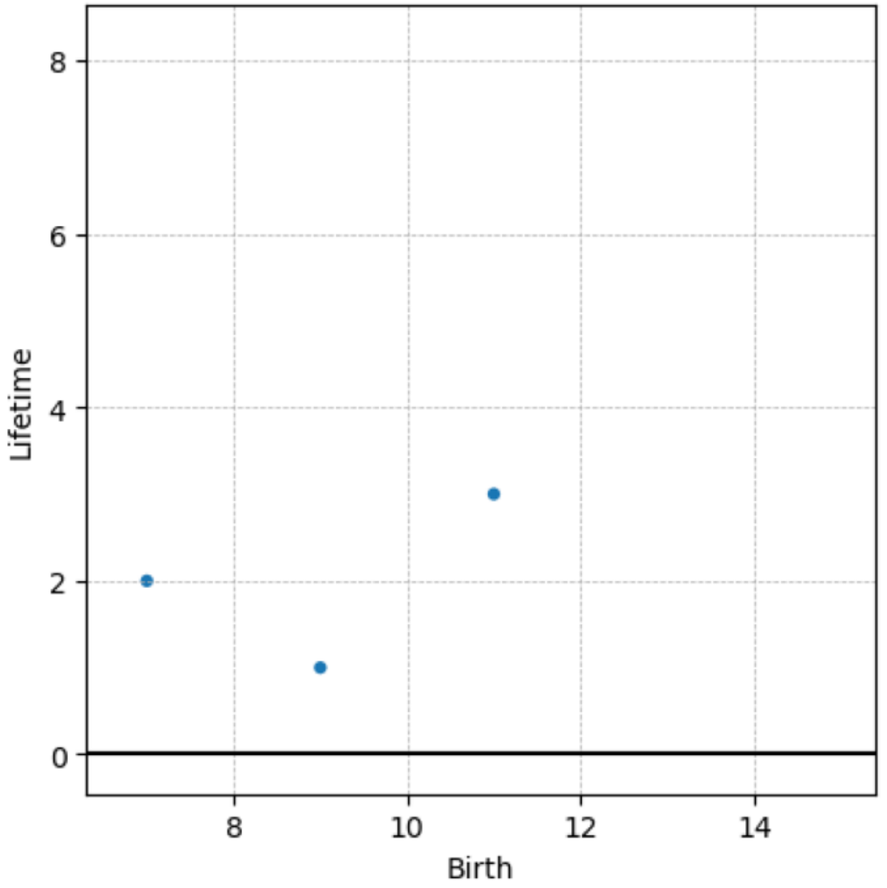
Persistence diagram



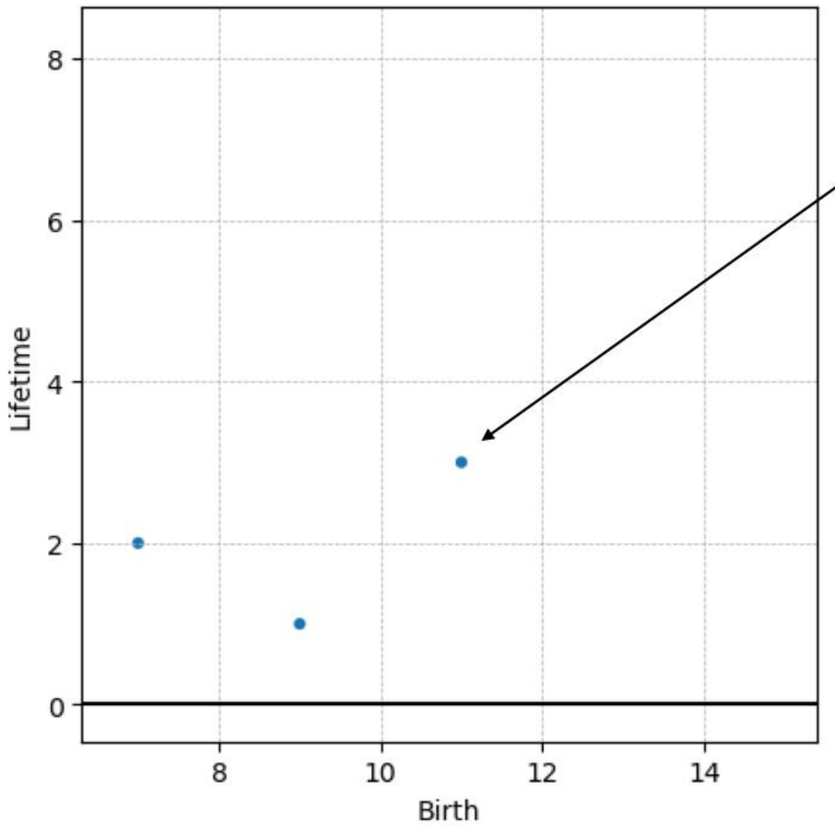
Persistence diagram



Lifetime diagram

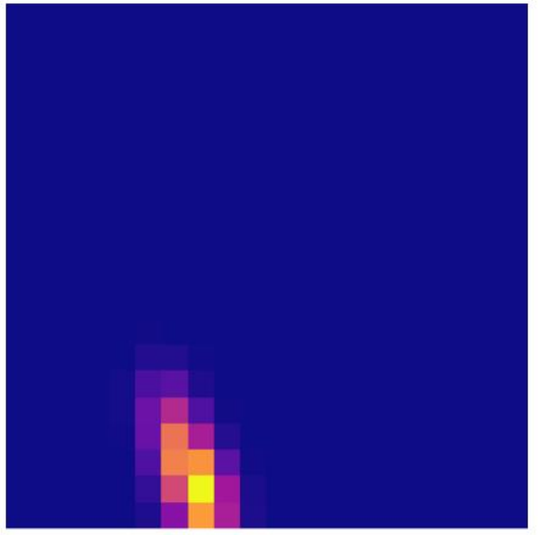


Lifetime diagram



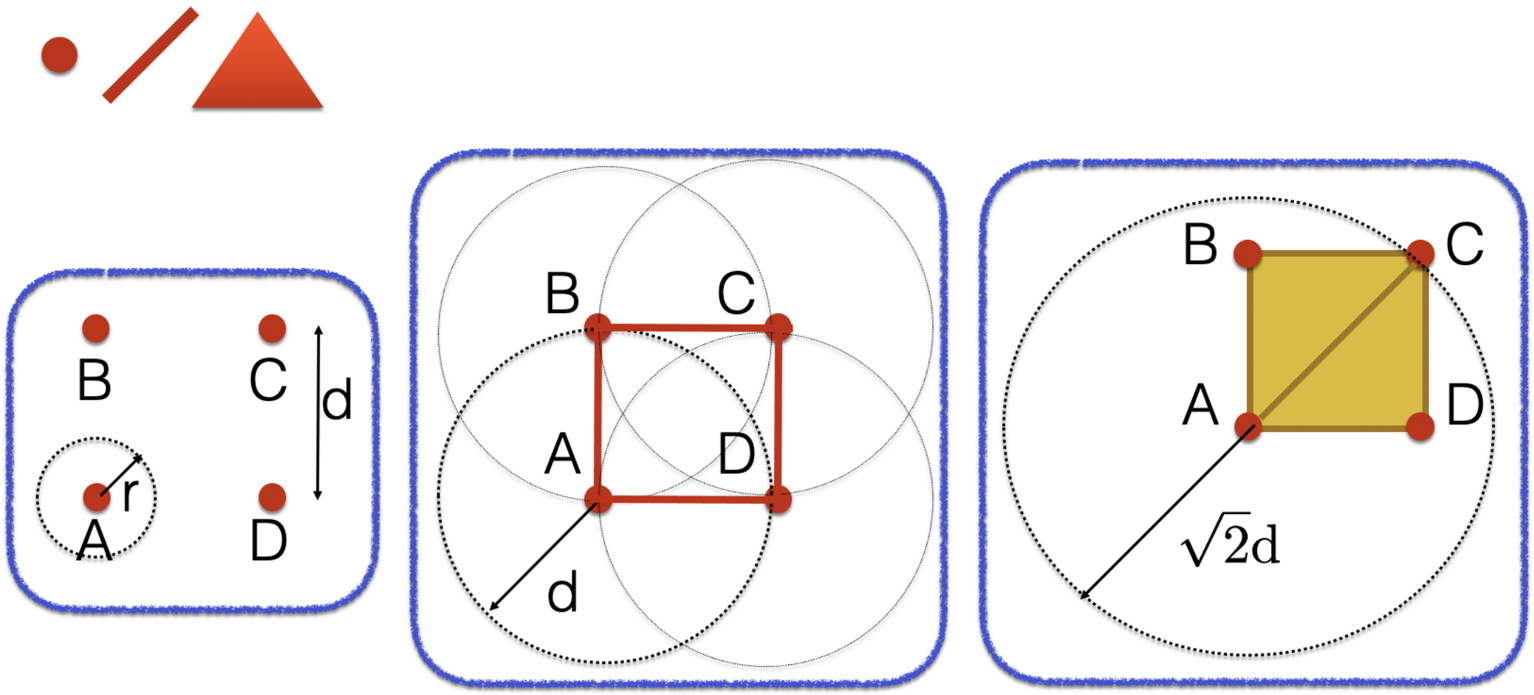
$$\rho_S(u,v) = \sum_{(x,y)\in\mathcal{T}(S)} w(x,y)\, g_{(x,y)}(u,v).$$

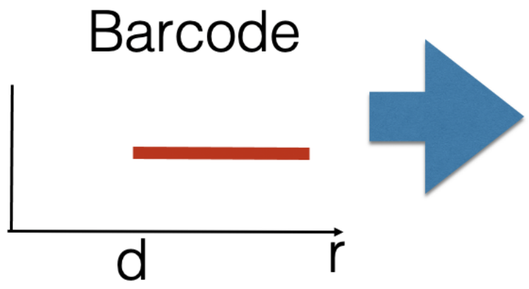
$$\mathcal{PI}_{P_i}(S) = \iint_{P_i} \rho_S(u,v)\, du\, dv.$$



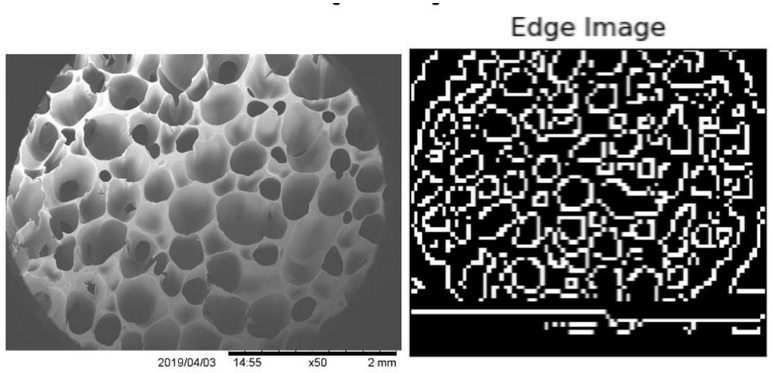
Persistence image

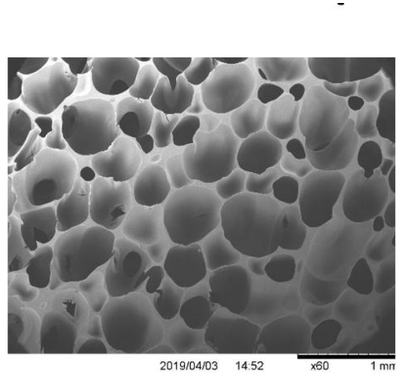
Images

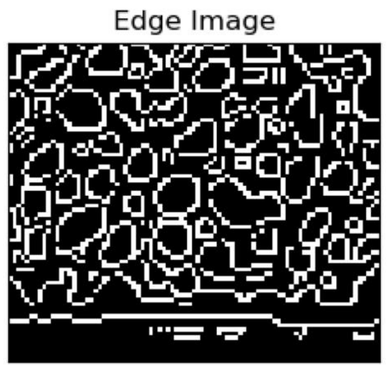


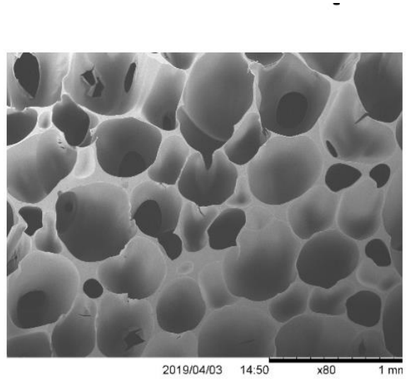


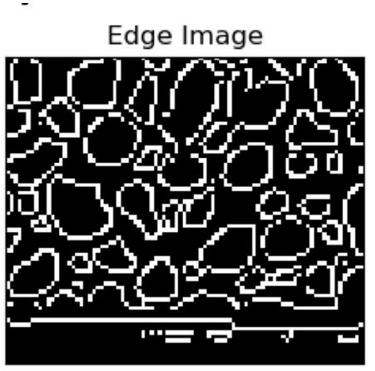
Persistence diagram,
Lifetime diagram
&
Peristence images

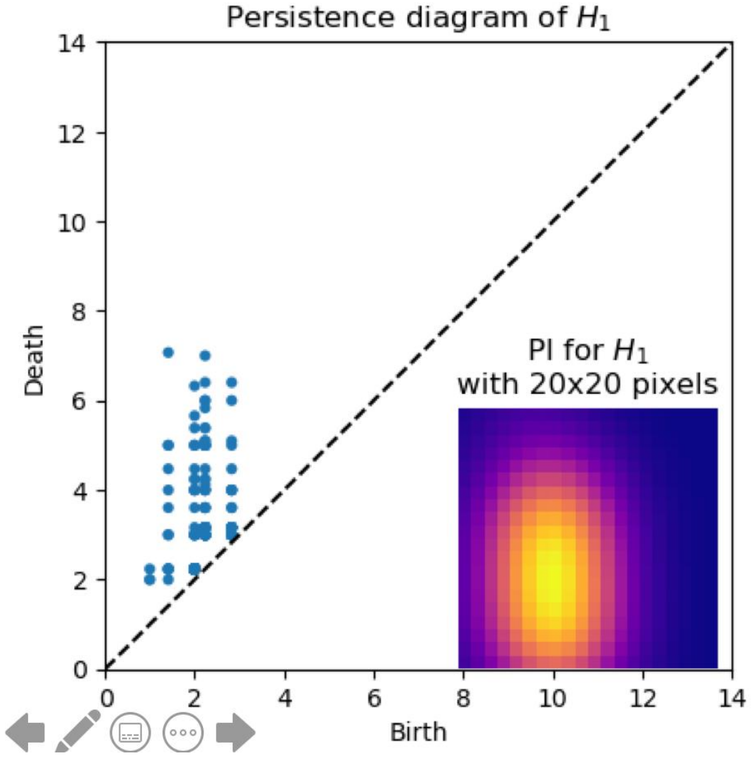

Edge Image
2019/04/03 14:55 x50 2 mm

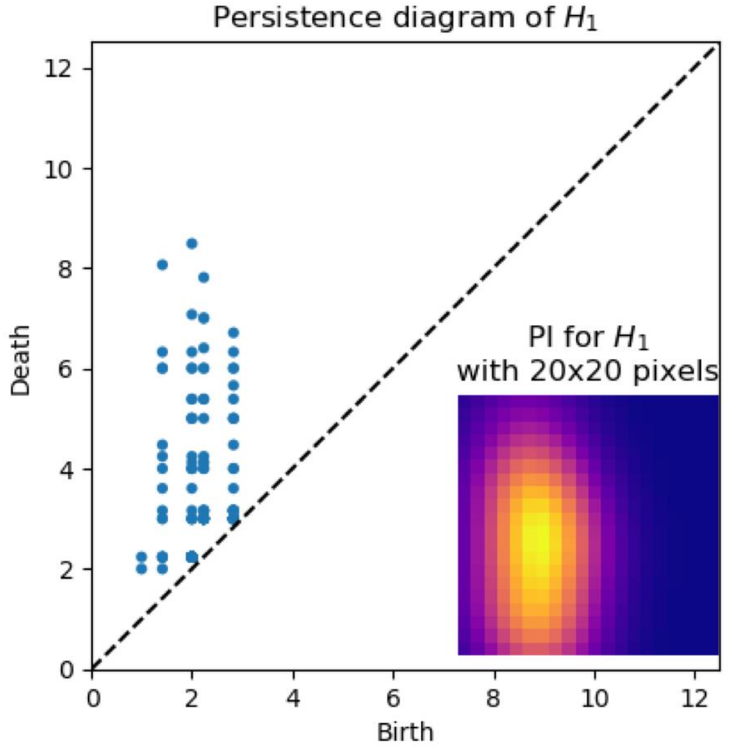

2019/04/03 14:52 x60 1 mm

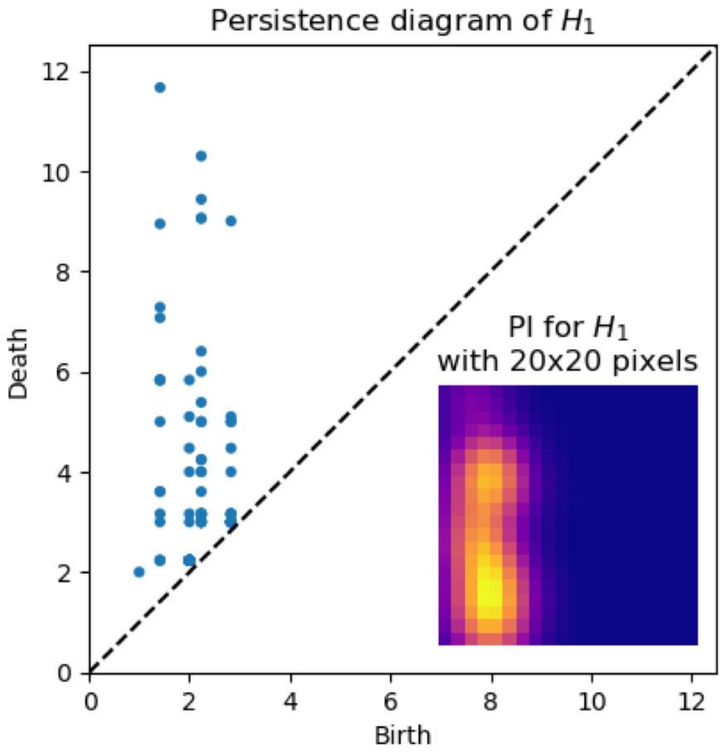

Edge Image


2019/04/03 14:50 x80 1 mm

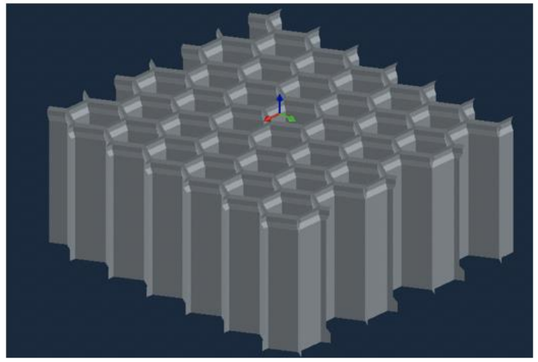

Edge Image

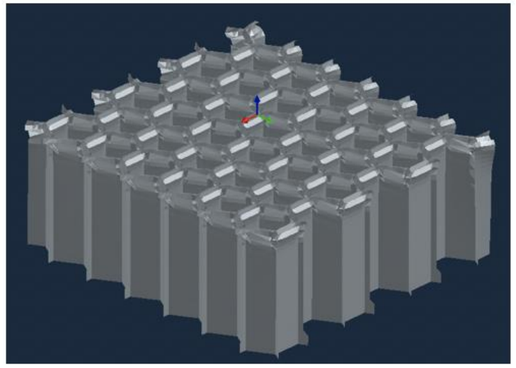

Persistence diagram of H_1
PI for H_1
with 20x20 pixels
Death
Birth
0
2
4
6
8
10
12
14

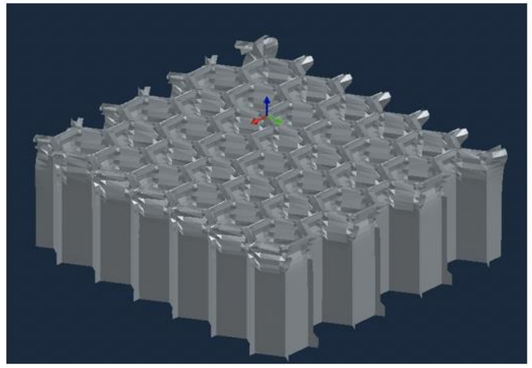

Persistence diagram of H_1
PI for H_1
with 20x20 pixels
Death
Birth
0
2
4
6
8
10
12

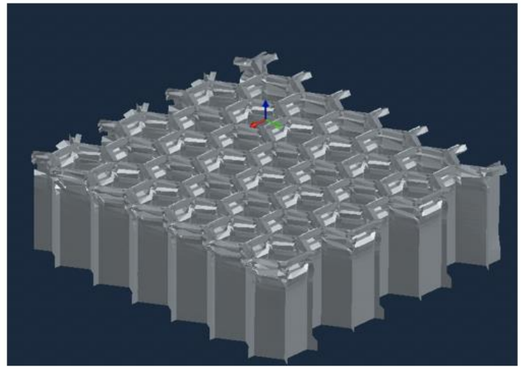

Persistence diagram of H_1
PI for H_1
with 20x20 pixels
Death
Birth
0
2
4
6
8
10
12

0s	0.85s	1.7s	2.55s

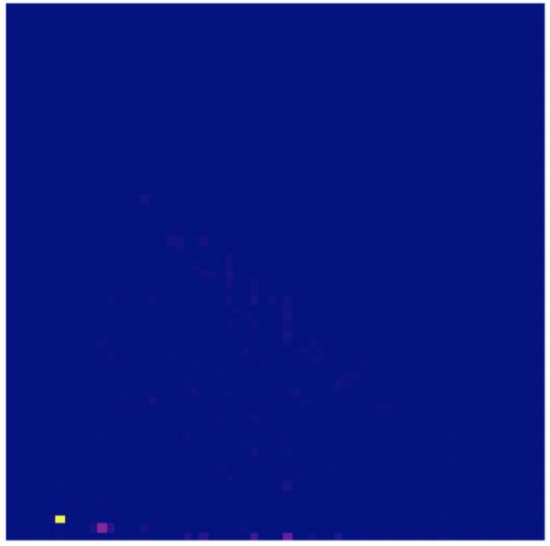




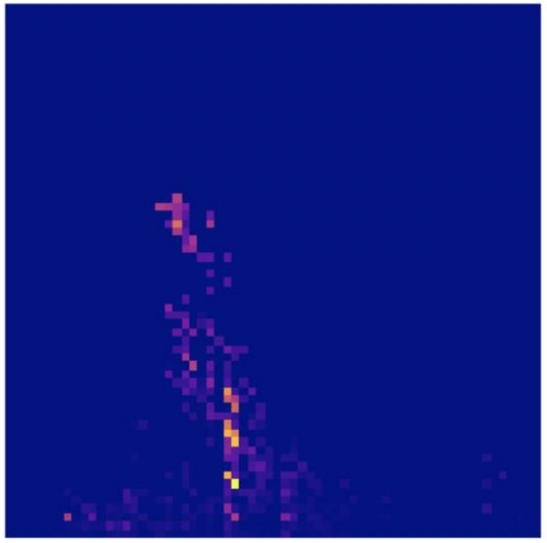




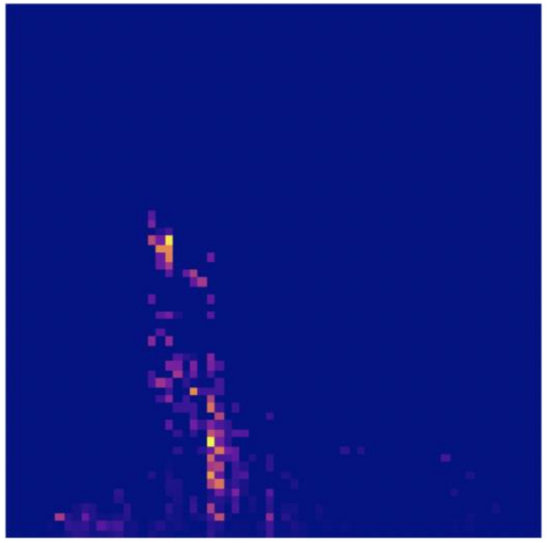
PI for H_1 with 64x64 pixels



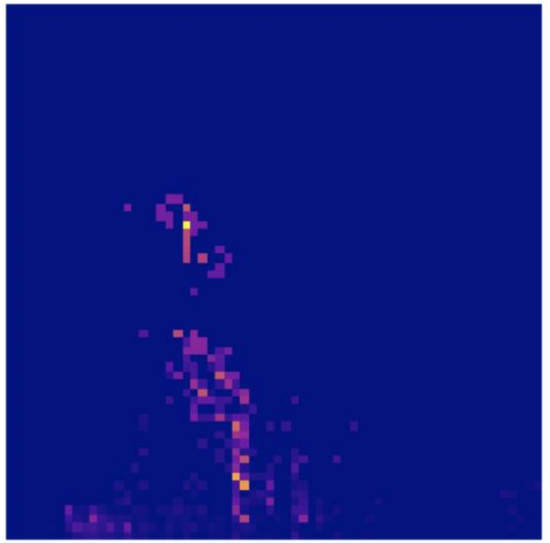
PI for H_1 with 64x64 pixels



PI for H_1 with 64x64 pixels



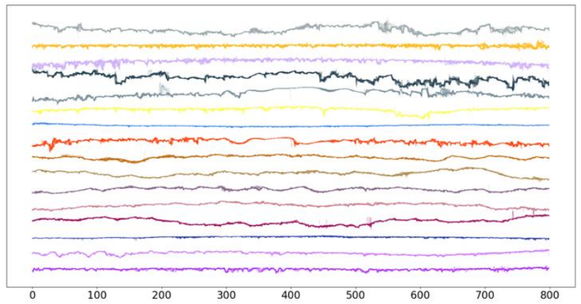
PI for H_1 with 64x64 pixels

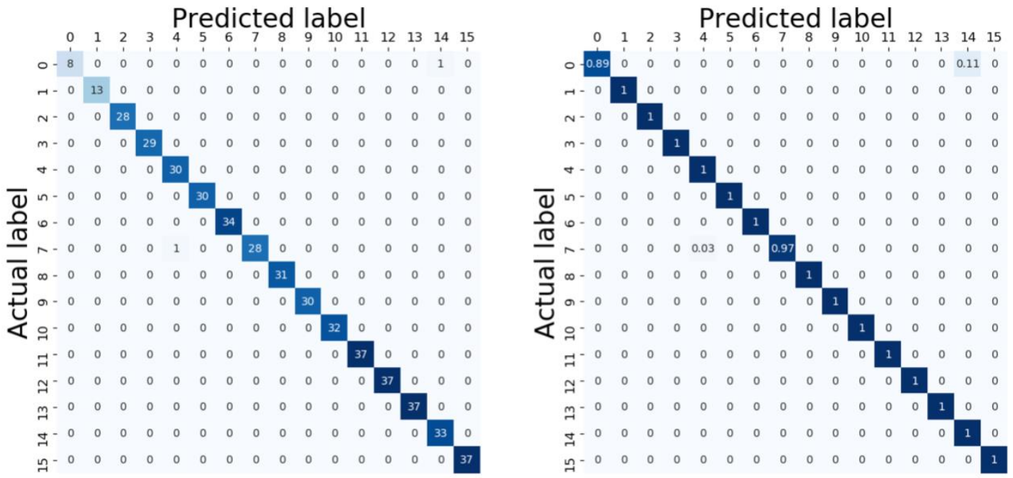


Applications

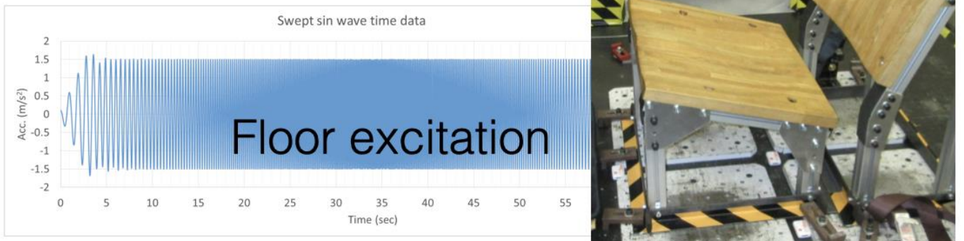
Rough surfaces clustering

16 families of composites

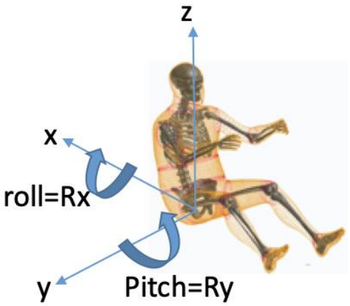




Human models







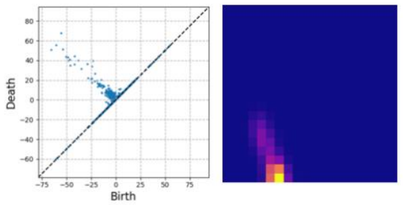


Tense, relax, …?

Data → **TDA**

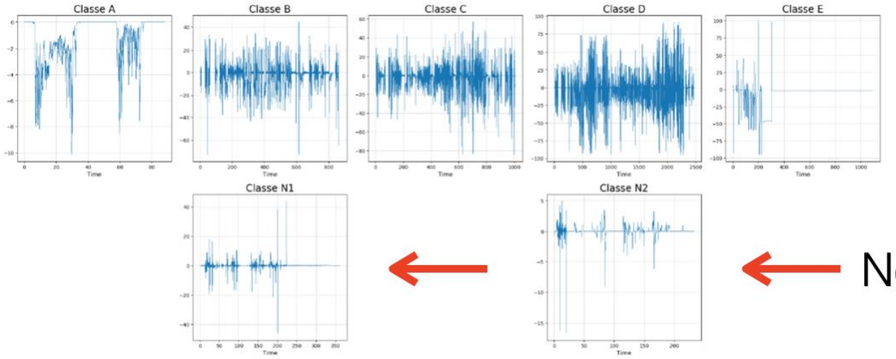


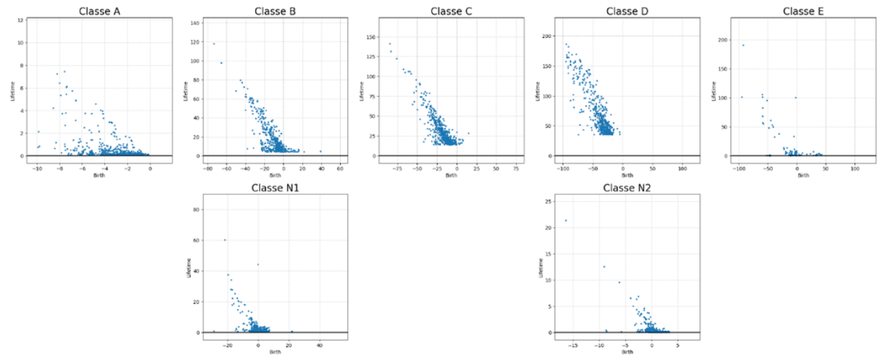
Barcode > Persistence diagram > persistence image > Behavioral classification

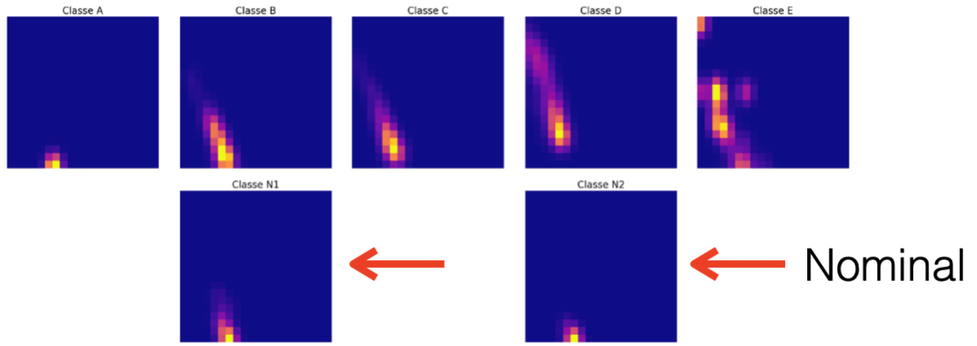


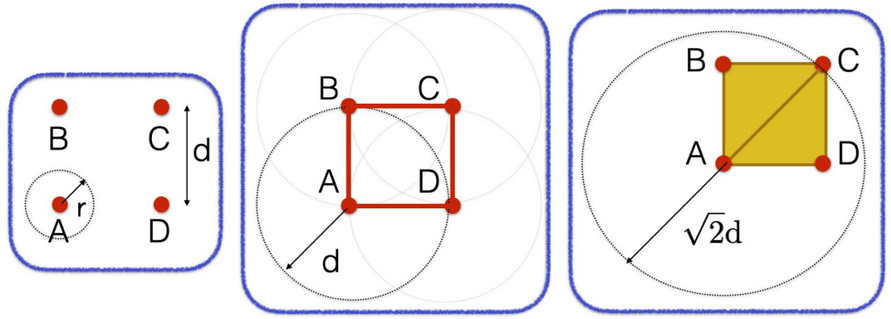
DIGITAL HUMAN LAB

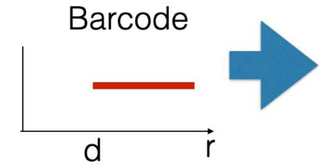
Obtained accuracy > 96%



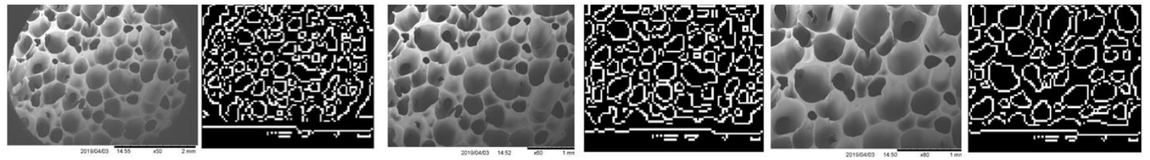


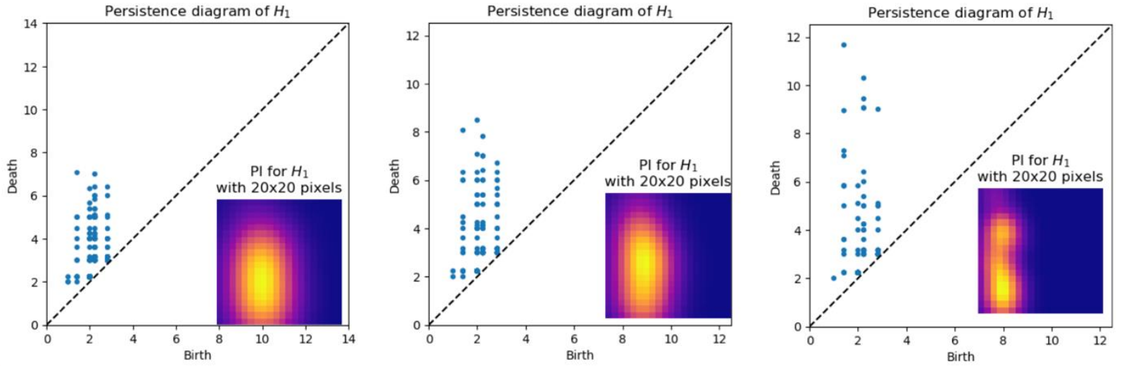




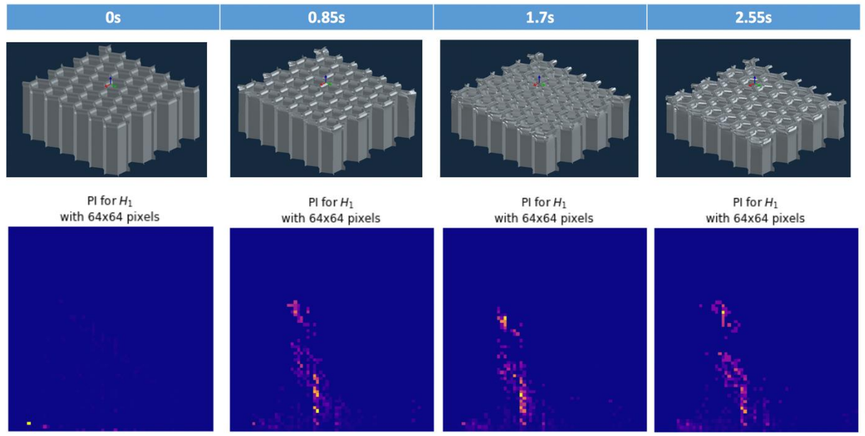


Persistence diagram,
Lifetime diagram
&
Peristence images

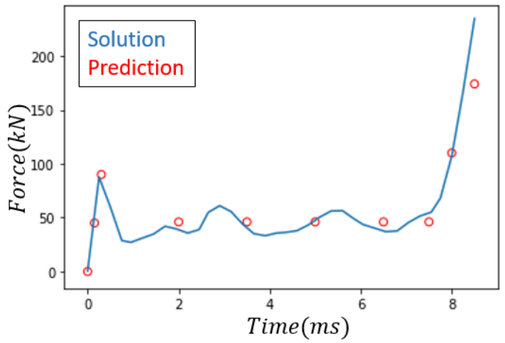




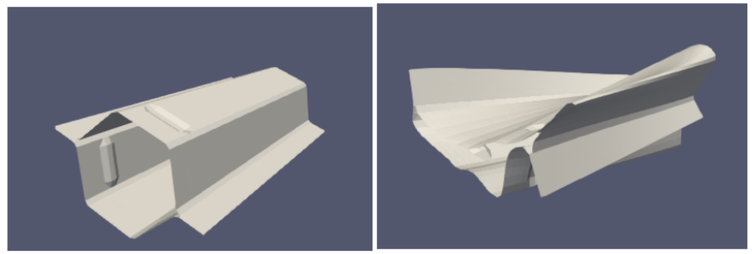
Data-driven modelling of honeycomb barriers

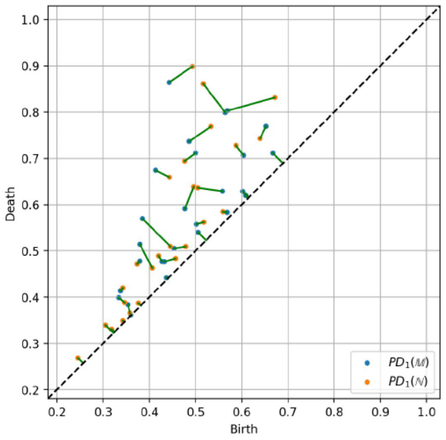


Neural Networks
or
Code2Vect
or
PGD



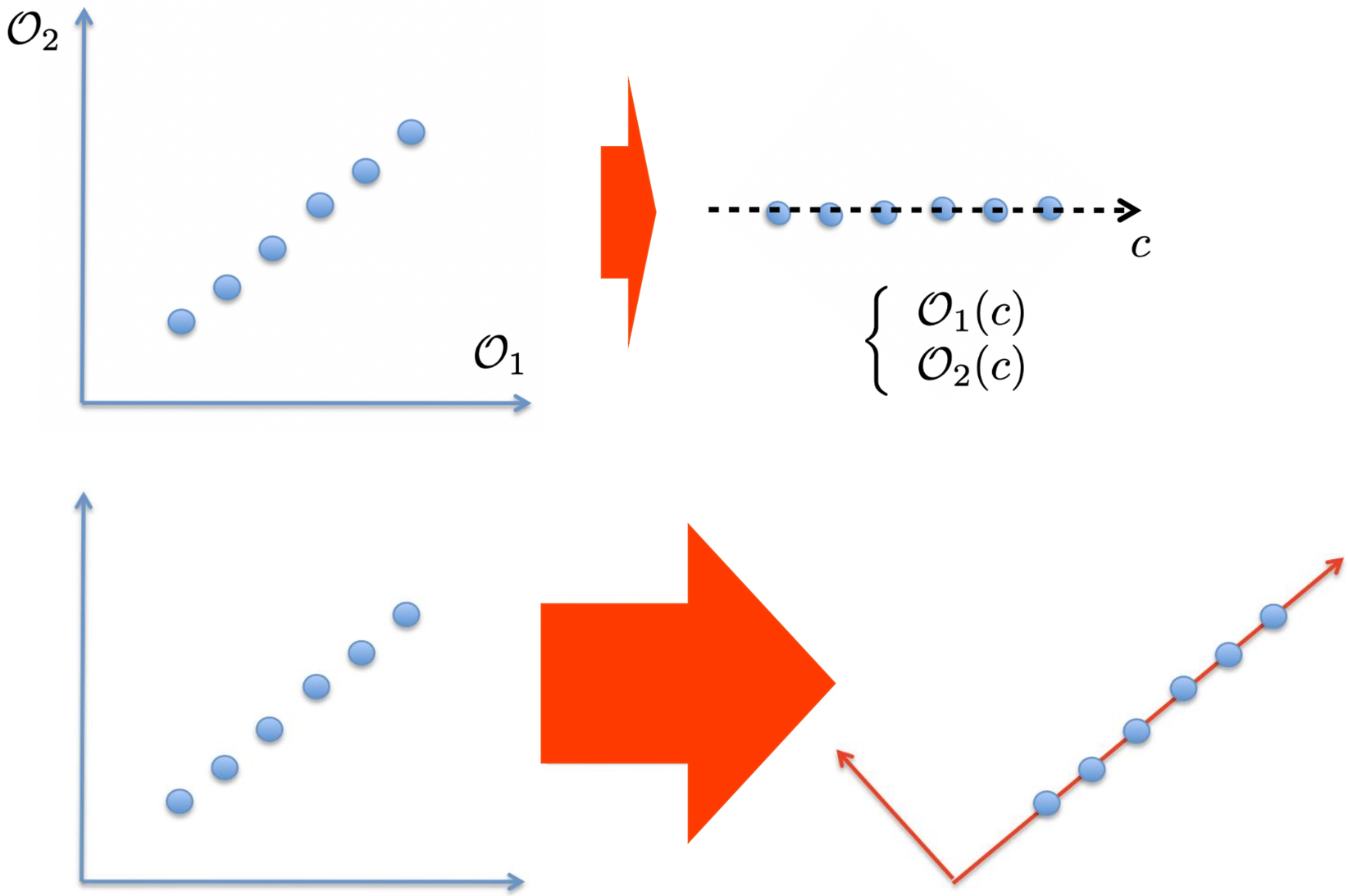
Parametric vibration modes clustering for advanced NVH





Case	1st surf.	2nd surf.	3rd surf.	4th surf.	5th surf.	6th surf.
01	1	2	3	4	5	6
02	1	2	3	4	5	6
03	1	2	3	4	5	6
04	1	2	3	5	4	6
05	1	2	3	4	5	6
06	1	2	3	5	4	6
07	1	2	3	4	5	6
08	1	2	3	4	5	6
09	1	2	4	3	6	5
10	2	1	3	4	6	5
11	1	2	3	4	5	6
12	1	2	3	4	5	6
13	1	2	3	5	4	6
14	1	2	4	3	6	5
15	1	2	4	3	5	6
16	1	2	3	4	5	6
17	1	2	3	4	6	5

(linear) Manifold Learning



Linear Manifold Learning

POD/PCA/KL/SVD

$$\mathbf{x}^i = (x_1^i, x_2^i, \cdots, x_N^i)^T$$

Data composed of ***N*** components for each parameters choice ***i***

$$\mathbf{Q} = (\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^M)$$

$$\texttt{Eig}(\mathbf{Q}\mathbf{Q}^T) \Rightarrow (\lambda_1, \mathbf{v}_1), \ldots, (\lambda_N, \mathbf{v}_N)$$

$$\lambda_{r+1} = \cdots = \lambda_N = 0$$

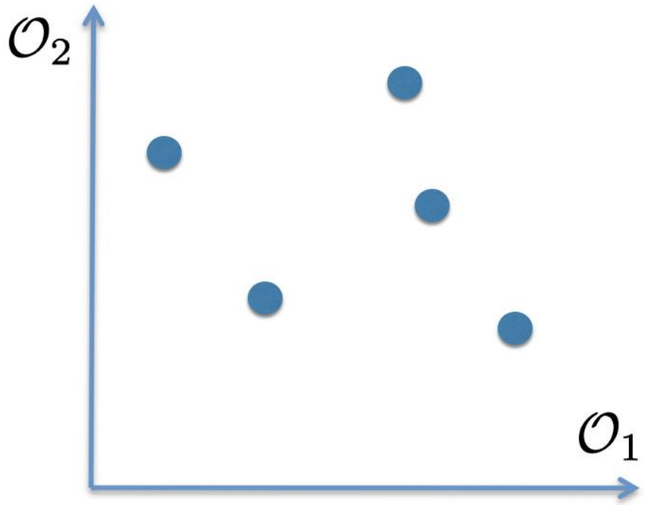
The prolem complexity, the solution manifold, has dimension ***r*** (***r***<<***N***)


AUX GRANDS HOMMES LA PATRIE RECONNAISSANTE

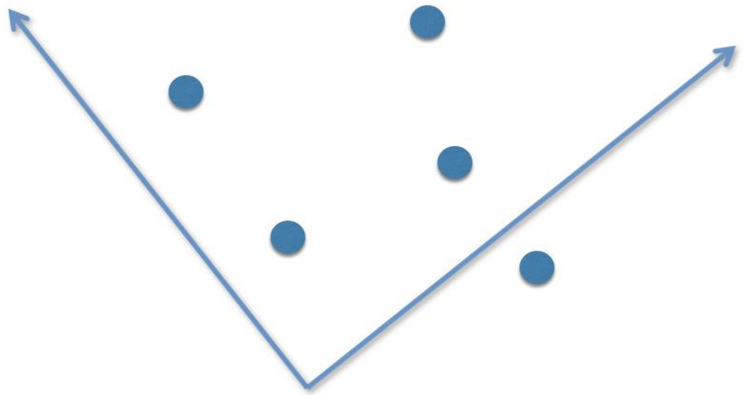
How many columns has the Pantheon front ?

22 !

Manifold Learning



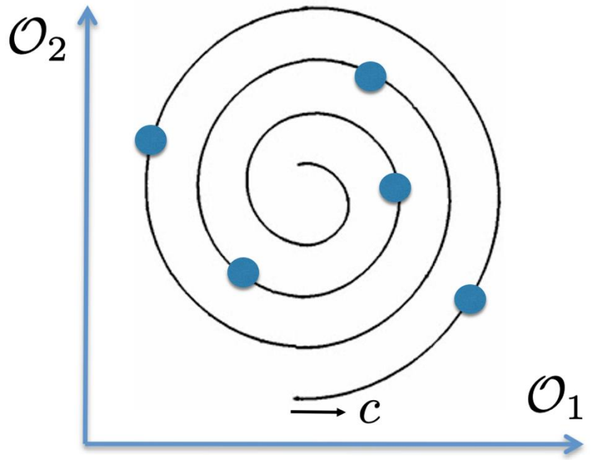
What is the dimensionality ?

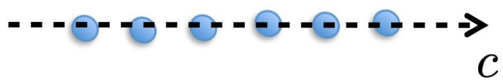


Apparently 2

Nonlinear Manifold Learning

However it is still 1 !

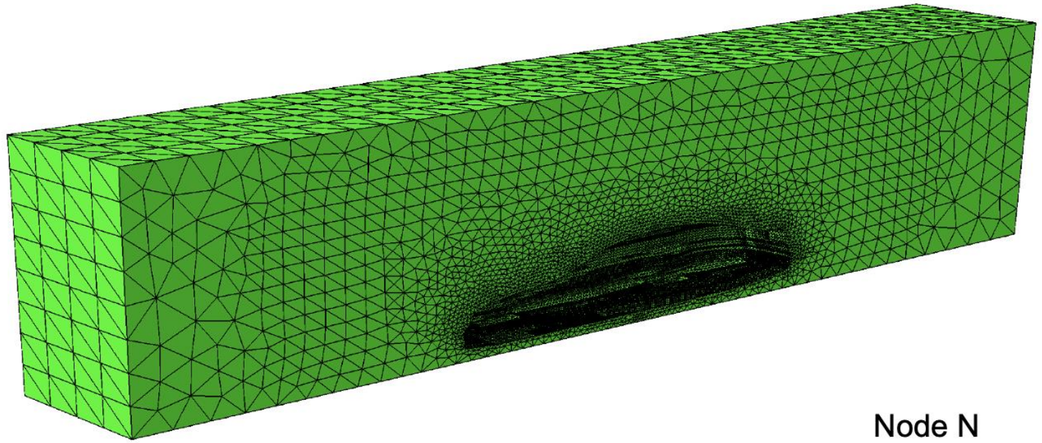




$$\begin{cases} \mathcal{O}_1(c) \\ \mathcal{O}_2(c) \end{cases}$$

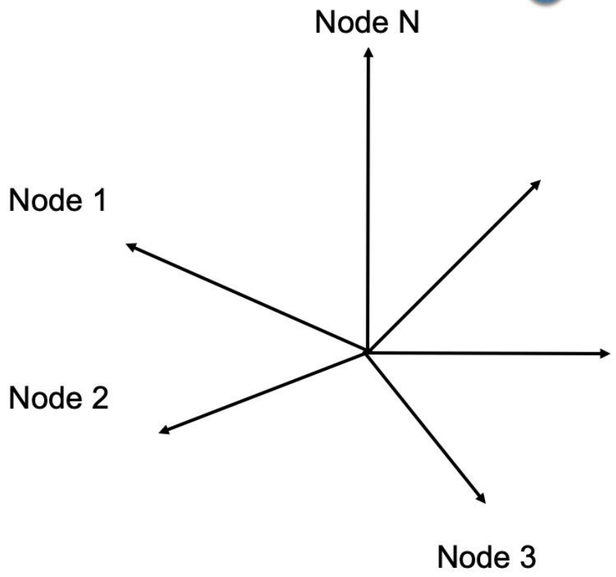
Linear model order reduction (e.g. PCA) fails whereas non-linear ones work (e.g. LLE, kPCA, IPCA,tSNE, …)

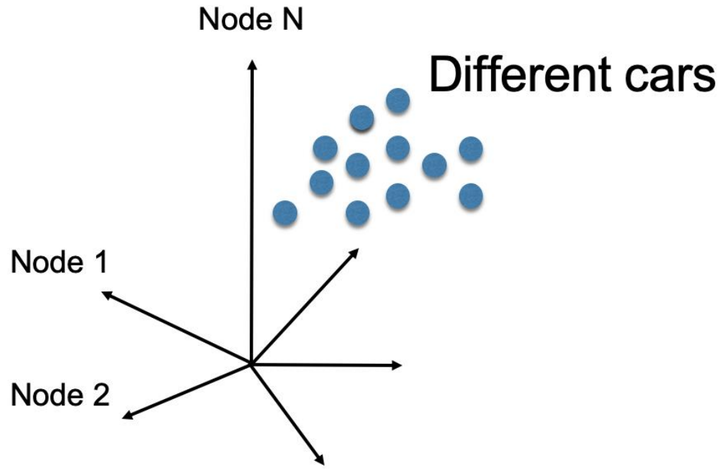
Nonlinear Manifold Learning for parametrizing shapes



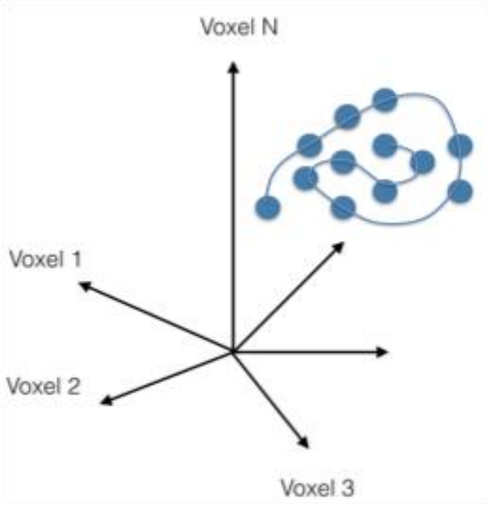
Level set: distance of each node to the car surface

Car 1 represented by its level-set function

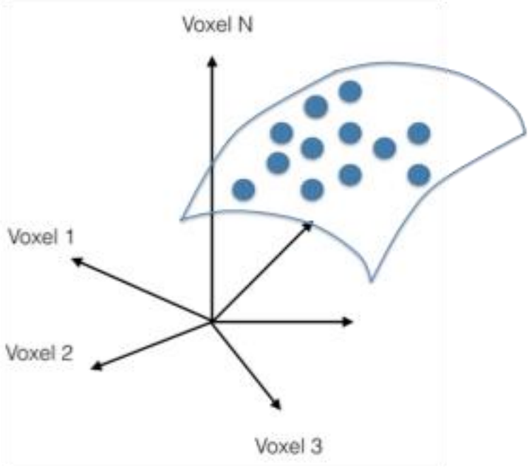




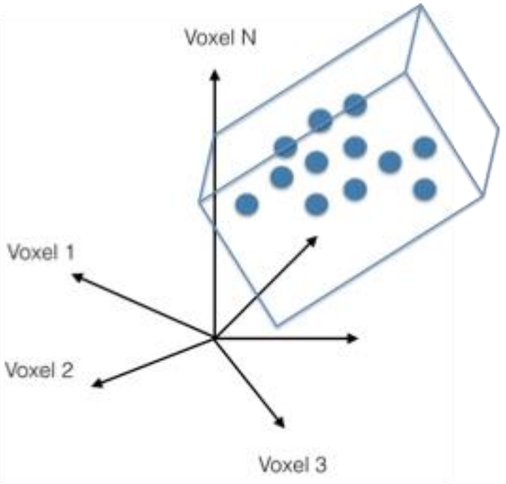
Nonlinear Manifold Learning



1D



2D

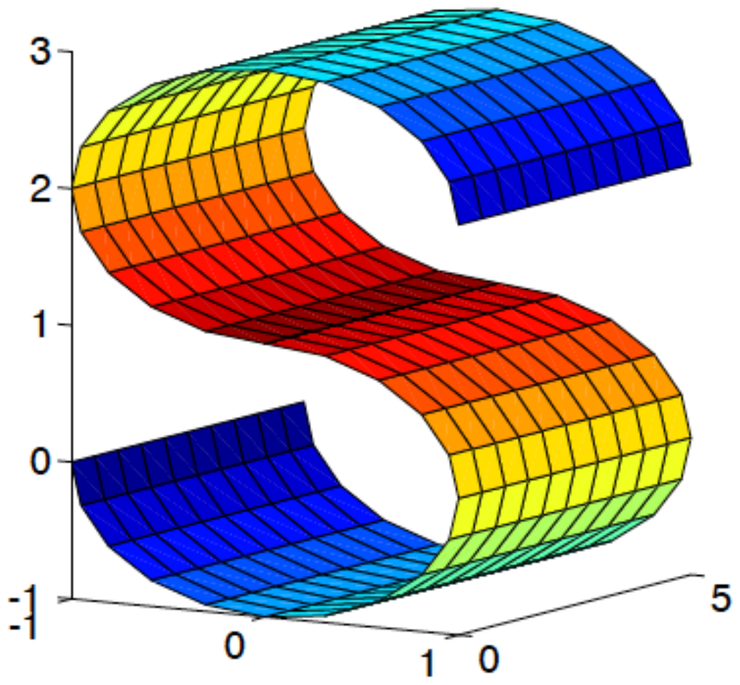


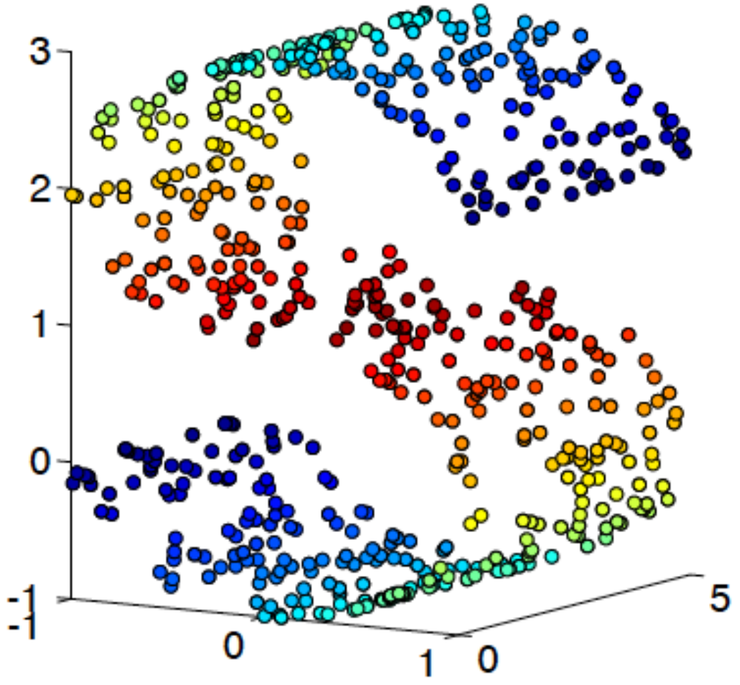
3D

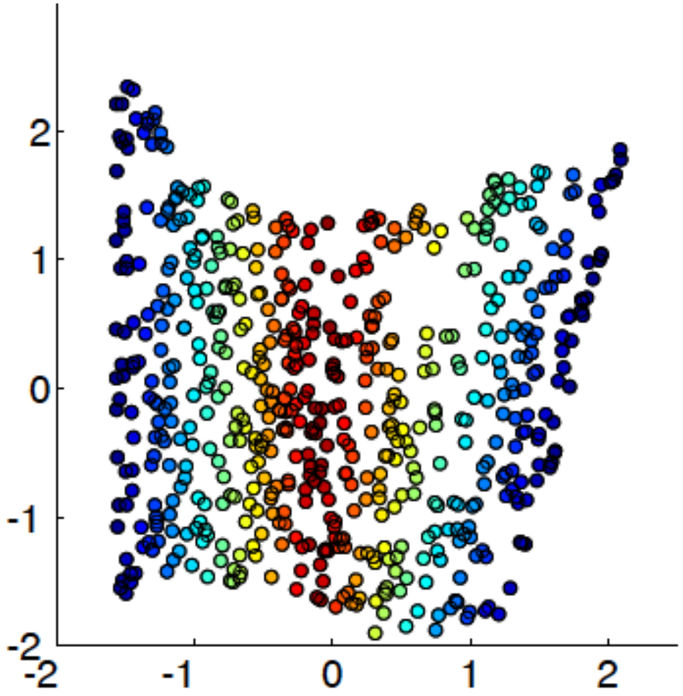
. . . . .

?

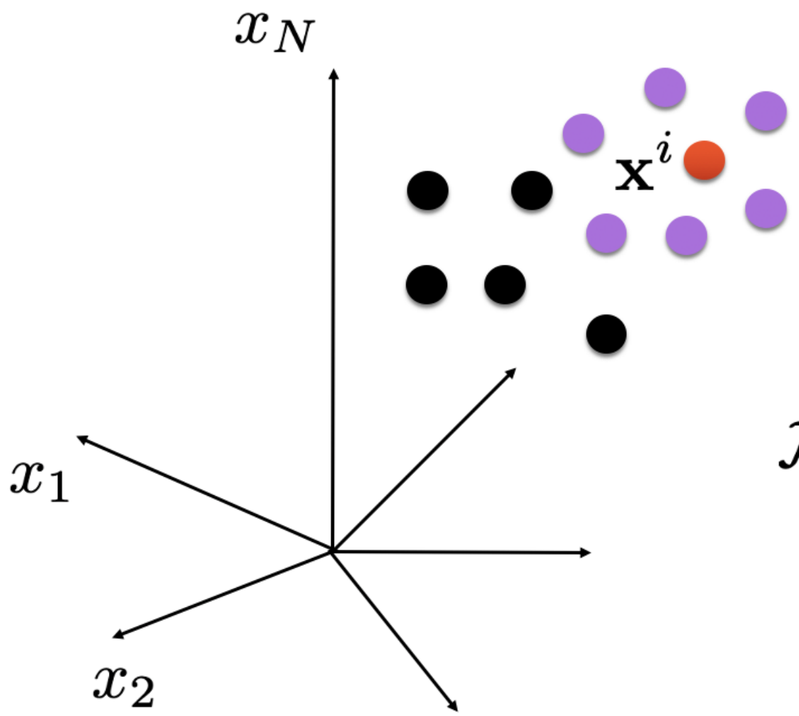
LLE







LLE

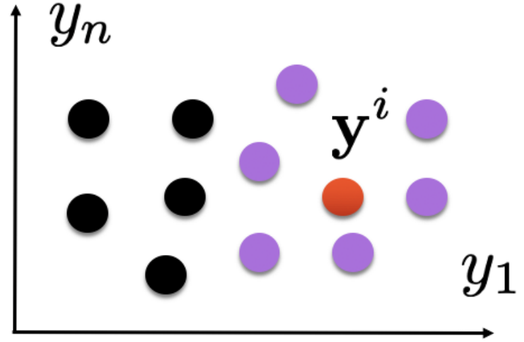


$$\mathbf{x}^i = \sum_{j \in \mathcal{V}_i} W_{ij} \mathbf{x}^j$$

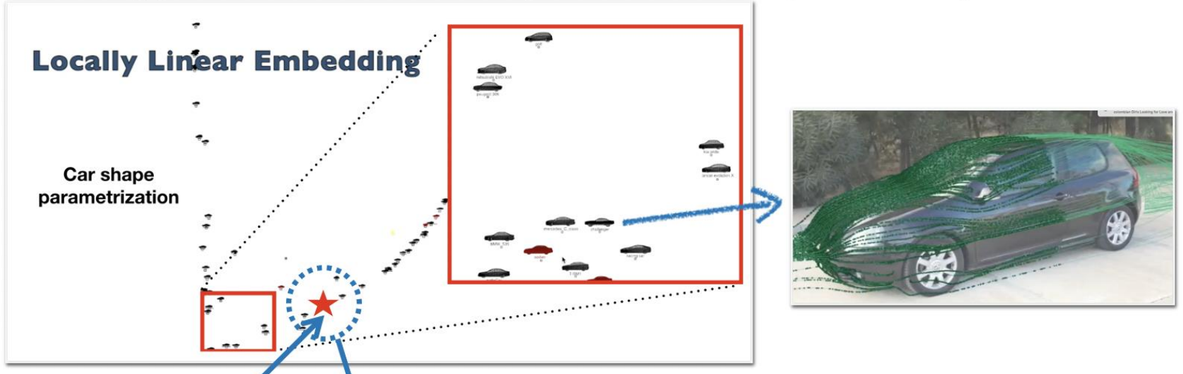
$$\mathcal{F}(W_{ij}) = \sum_i \left\{ \mathbf{x}^i - \sum_{j \in \mathcal{V}_i} W_{ij} \mathbf{x}^j \right\}^2$$

$$\mathbf{y}^i = \sum_{j \in \mathcal{V}_i} W_{ij} \mathbf{y}^j$$

$$\mathcal{G}(\mathbf{y}^i) = \sum_i \left\{ \mathbf{y}^i - \sum_{j \in \mathcal{V}_i} W_{ij} \mathbf{y}^j \right\}^2$$



How many parameters define car shapes from the aerodynamic viewpoint?



New car NEVER SIMULATED

CFD solution interpolated on the manifold from its neighbors



Augmented Reality using High-Fidelity Models

Wind Tunnel

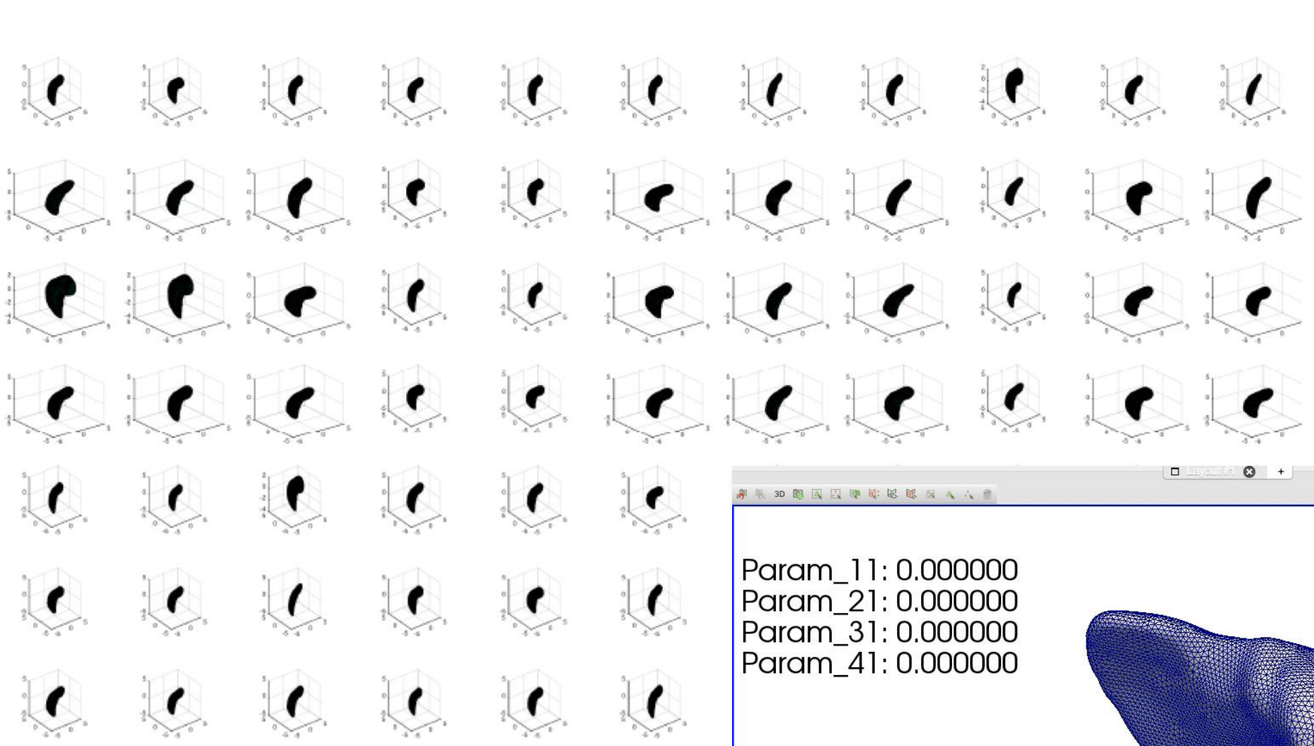
A. Badías, I. Alfaro, D. González, F. Chinesta and E. Cueto

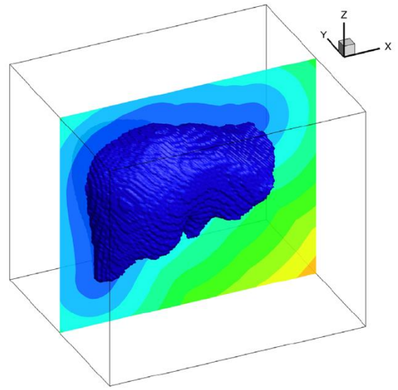
abadias@unizar.es

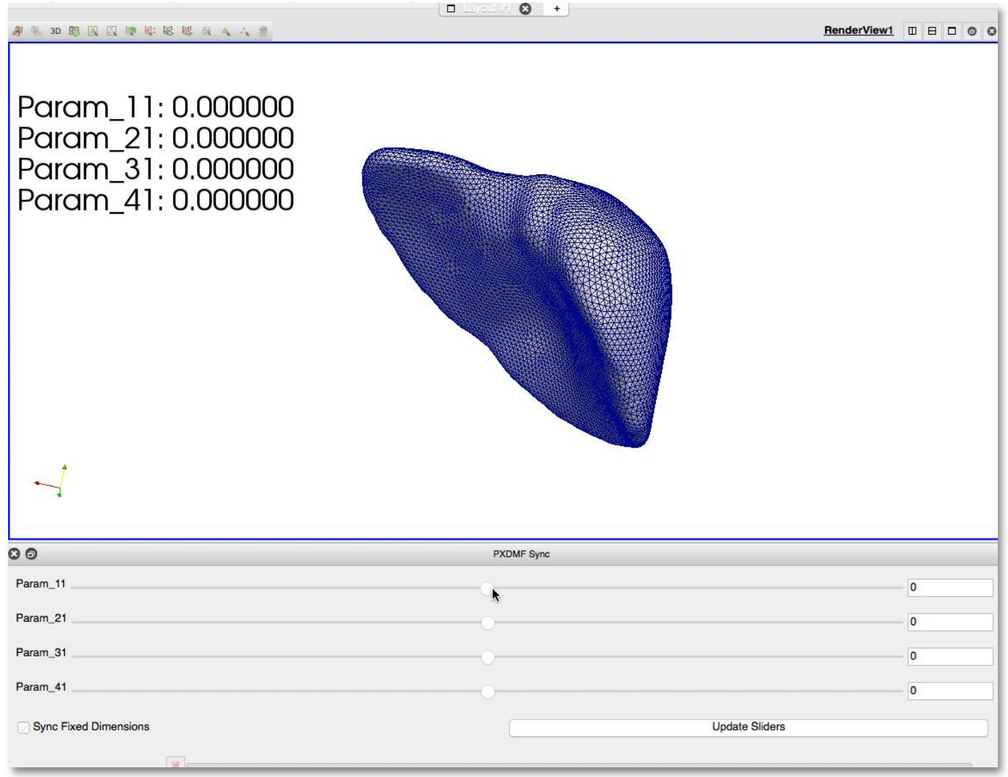
amb applied mechanics & bioengineering

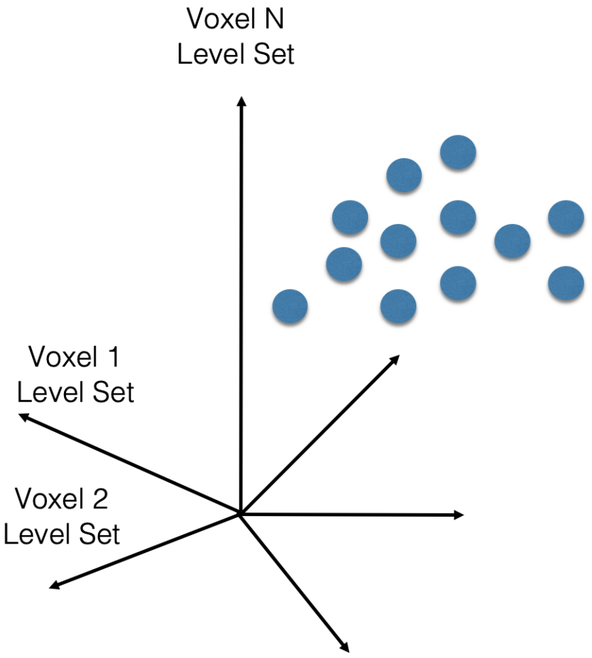
Instituto Universitario de Investigación en Ingeniería de Aragón Universidad Zaragoza

Universidad Zaragoza 1542




Z
Y
X


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Sync Fixed Dimensions
Update Sliders


Voxel N
Level Set
Voxel 1
Level Set
Voxel 2
Level Set